



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

MECHANICAL ENGINEERING (Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr.	L	T	P	Internal Marks	External Marks	Total Marks
B19 HS 3102	Operations Research	HS	3	3	--	--	25	75	100
B19ME 3101	IC Engines & Gas Turbines	PC	3	3	--	--	25	75	100
B19 ME 3102	Kinematics of Machines	PC	3	3	--	--	25	75	100
B19 ME 3103	Design of Machine Elements	PC	3	3	--	--	25	75	100
B19 ME 3104	Fluid Machines & Systems	PC	3	3	--	--	25	75	100
B19 ME 3105	Mechanical Measurements & Metrology	PC	3	3	--	--	25	75	100
B19 ME 3106	IC Engines & Kinematics of Machines Lab	PC	1.5	--	--	3	20	30	50
B19 ME 3107	Metrology Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability Skills-I	MC	--	3	--	--	--	--	--
TOTAL			21	21	--	6	190	510	700

Code	Category	L	T	P	C	I.M	E.M	Exam
B19HS3102	HS	3	--	--	3	25	75	3 Hrs.
OPERATIONS RESEARCH								
(For ME)								
Course Objectives:								
1.	To acquaint the students with basic Operation Research concepts, Formulation of LPP and its solution using various methods.							
2.	To build capabilities in the students to analyse the various transportation and assignment problems, job sequencing problems, inventory problems, Games theory and Queuing Models.							
3.	To familiarise the students with project management techniques i.e., PERT and CPM .							
Course Outcomes: Upon successful completion of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Describe the basic Operations Research models, formulate and solve Linear Programming problems for industrial and business applications							K4
2.	Build and Solve Transportation and Assignment problems using appropriate methods for different situations.							K4
3.	Determine the optimal solutions for various Job Sequencing and Inventory models for industrial applications.							K4
4.	Analyse and solve various Games theory and Queuing Models in real situations							K4
5.	Design and schedule various project management problems by CPM & PERT.							K4
SYLLABUS								
UNIT-I (10Hrs)	Introduction to OR: Definition of OR, Characteristics and phases of OR, Scope of OR, OR models, General methods for solving OR models, Roll of computers in OR. Linear Programming: Formulation, Graphical Solution, Simplex Method, Artificial Variable Technique-Big-M method, Duality.							
UNIT-II (10 Hrs)	Transportation Model: Balanced and Unbalanced transportation problems-Initial solution by North West Corner Rule, Lowest Cost Method and VAM, Optimality test by MODI method, Degeneracy in TP. Assignment Model: Hungarian algorithm, Balanced and Unbalanced Assignment Problems, Travelling Salesman Problems.							
UNIT-III (10 Hrs)	Job Sequencing: Introduction, Assumptions, Johnson's algorithm for N-Jobs 2-Machines Problems, N-Jobs 3-Machines Problems, N-Jobs M-Machines Problems, Graphical solution for 2-Jobs and M-Machines Problems. Inventory Models: Definition of Inventory, Costs associated with Inventory Problems, Classification of Models, EOQ Model with and without Shortages, Inventory Problems with Price Breakups.							
UNIT-IV (10 Hrs)	Game Theory: Introduction, Basic definitions, Two Person Zero Sum Games, Minimax criterion, Saddle point, Value of game, Solution of games with saddle point, Mixed Strategy Games-Arithmetic method, Dominance principle to reduce size of game, Graphical Method, Algebraic solution to rectangular games. Queuing Theory: Structure of Queuing Models, Characteristics of Queuing process, Kendall's notation, Single channel systems-(M/M/1:∞/FIFO) model and (M/M/1:N/FIFO) model.							

UNIT-V (10 Hrs)	Network Analysis: Introduction, Project scheduling by CPM and PERT, Network diagram representations, Rules to construct Network diagrams, Time estimates in network analysis- EST, EFT, LST, LFT, float/slack and critical path, Time estimates and Probability considerations in PERT, Crashing in PERT
Text Books:	
1.	Operations Research by S.D Sharma.
2.	Operations Research by V. K. Kapoor.
Reference Books:	
1.	Operations Research - Kanti Swaroop, P.K. Gupta, Man Mohan, Sulthan Chand&Sons Education.
2.	Operations Research - Hamdy A Taha – Pearson Education.
3.	Operations Research -Panneer Selvan Prentice Hall of India.
4.	Introduction to Operations Research, F.S. Hiller, G.J. Liberman, TMH.
Web links	
1.	https://nptel.ac.in/courses/112/106/112106134/
2.	https://nptel.ac.in/courses/110/106/110106062/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3101	PC	3	--	--	3	25	75	3 Hrs.
IC ENGINES & GAS TURBINES								
(For ME)								
Course Objectives:								
1.	To impart the knowledge of basic gas power cycles and combustion phenomena, various performance factors and design aspects of SI and CI Engines.							
2.	To impart the knowledge of combustion phenomenon taking place in SI engines							
3.	To impart the knowledge of combustion phenomenon taking place in CI engines							
4.	To impart the knowledge of Reciprocating compressors and Rotary compressors methods to improve their efficiency.							
5.	To impart the knowledge of gas turbine cycles, and their components and efficiency improving techniques, and jet propulsion.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the knowledge of gas power cycles adequately and can calculate their efficiencies.							K3
2.	Explain the processes involved in combustion in S.I Engines.							K3
3.	Explain the processes involved in combustion in C.I Engines.							K3
4.	Apply the knowledge of reciprocating and rotary compressors in engineering applications.							K3
5.	Compute and develop various methods to improve the efficiency of gas turbine power plants, and can explain jet propulsions.							K3
SYLLABUS								
UNIT-I (10 Hrs)	I.C. engines: classification-comparison of two strokes and four stroke engines- comparison of S.I. and C.I. engines-Valve timing and port timing diagrams- Efficiencies- air standard efficiency indicated thermal efficiency, brake thermal efficiency, mechanical efficiency, volumetric efficiency and relative efficiency-Testing and performances of I.C. engines - Basic principles of carburetion and fuel injection.							
UNIT-II (10 Hrs)	Combustion in S.I. Engines- Normal combustion and abnormal combustion Importance of flame speed and effect of engine variables-types of abnormal combustion pre-ignition and knock, Fuel requirements and fuel rating, anti-knock additions- Combustion chamber requirements and Types of combustion chamber- Design principles of combustion chambers.							
UNIT-III (10 Hrs)	Combustion in C.I. Engines- Stages of combustion- Delay period and its importance- effect of engine variables, diesel knock, suction compression and combustion induced turbulence, open and divided combustion chambers.							
UNIT-IV (10 Hrs)	Reciprocating and Rotary Compressors: Reciprocating compressors-effect of clearance in compressors, volumetric efficiency-single stage and multi stage compressors-effect of inter cooling in multi stage compressors-Vane type blower-centrifugal compressor- Adiabatic efficiency- Diffuser- Axial flow compressors- Velocity diagrams, degree of reaction, performance characteristics.							

UNIT-V (10 Hrs)	Gas Turbines: Simple gas turbine plant- Ideal cycle, closed cycle and open cycle for gas turbines Efficiency, work ratio and optimum pressure ratio for simple gas turbine cycle- Parameters of performance- Actual cycle, regeneration, Inter-cooling and reheating, closed and semi-closed cycle Jet propulsion and Rockets, Applications of Gas Turbines.
Text Books:	
1.	A Treatise on Heat Engineering by Vasandhani and Kumar.
2.	Applied Thermodynamics-II by R. Yadav
Reference Books:	
1.	I.C. Engines by V. Ganesan.
2.	Thermal Engineering, by R.K.Rajput.
3.	I.C. Engines by Mathur and Mehata.
4.	Gas Turbines, by Cohen and Rogers.
Web links	
1.	https://nptel.ac.in/courses/112/103/112103262/#
2.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3102	PC	3	--	--	3	25	75	3 Hrs.
KINEMATICS OF MACHINES								
(For ME)								
Course Objectives:								
1.	To Impart the knowledge about various plane mechanisms and its inversions							
2.	To provide foundation for relative motion, velocity and accelerations of the various elements in a mechanism.							
3.	To understand the concept of various straight line motion mechanisms and basic steering gears							
4.	To facilitate students to understand the functions of cams, gears and gear trains							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Examine the plane motion mechanism with single degree of freedom							K3
2.	Illustrate the importance of relative motion, velocity, and accelerations of the various elements in a plane mechanism							K3
3.	Analyze various plane mechanisms like straight line motion mechanisms and Automobile steering gear mechanisms							K4
4.	Understand the importance of various power transmission systems and analyze motion of higher pairs like toothed gears, gear trains and cams							K4
SYLLABUS								
UNIT-I (10 Hrs)		Mechanisms: Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematics pairs – sliding, turning, rolling, screw and spherical pairs – lower and higher pairs – closed and open pairs – constrained motion – completely, partially or successfully and incompletely constrained . Mechanism and Machines –Mobility of Mechanisms : Degree of freedom, Grubler’s criterion, classification of machines – kinematics chain – inversions of mechanism – inversions of four bar chain, single and double slider crank chains, Mechanical Advantage and Transmission angle						
UNIT-II (10 Hrs)		Kinematics: Velocity and acceleration – Motion of a link – Determination of Velocity and acceleration – Graphical method – Application of relative velocity method. Plane motion of body: Instantaneous center of rotation– Three centers in line theorem – Graphical determination of instantaneous center, determination of angular velocity of links by instantaneous center method. Analysis of Mechanisms: Analysis of slider crank chain for displacement- velocity and acceleration of slider – Acceleration diagram for a given mechanism. Coriolis acceleration - determination of Coriolis component of acceleration						

UNIT-III (10 Hrs)	Straight-line motion mechanisms: Exact and approximate copied and generated types –Peaucellier - Hart - Scott Russel- Modified Scott Russel – Grasshopper – Watt - Tchebicheff's - Pantographs Steering gears: Conditions for correct steering – Davis Steering gear, Ackerman's steering gear. Hooke's Joint: Single and double Hooke's joint –velocity ratio – application – problems
UNIT-IV (10 Hrs)	Cams: Definitions of cam and followers – their uses – Types of followers and cams – Terminology – Types of follower motion - Uniform velocity, Simple harmonic motion and uniform acceleration and retardation. Construction of cam profiles- Cam with knife edged follower and roller follower Maximum velocity and maximum acceleration during outward and return strokes.
UNIT-V (10 Hrs)	Higher pair: Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion – involutes profiles – phenomena of interferences – Methods of interference. Condition for minimum number of teeth to avoid interference – expressions for arc of contact and path of contact of Pinion & Gear Gear Trains: Introduction – Types – Simple – compound and reverted gear trains – Epicyclic gear train. Methods of finding train value or velocity ratio of Epicyclic gear trains. Torques in epicyclic gear trains
Text Books:	
1.	Theory of Machines / S.S Ratan/ Mc. Graw Hill Publ
2.	Theory of machines / Khurmi / S.Chand.
3.	Mechanism and machine theory by Ashok G. Ambedkar, PHI Publications
Reference Books:	
1.	Mechanism and Machine Theory / JS Rao and RV Duggipati / New Age.
2.	Theory of Machines / Shigley / MGH
3.	Theory of Machines / Thomas Bevan / CBS Publishers

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3103	PC	3	--	--	3	25	75	3 Hrs.
DESIGN OF MACHINE ELEMENTS								
(For ME)								
Course Objectives:								
1.	Introduce the concepts of design philosophy at basic level, engineering materials, types of loading and different failure criteria.							
2.	Acquainted with standards like ASTM, ASME etc., safety, reliability, importance of dimensional parameters, manufacturing aspects in mechanical design.							
3.	Acquaint with the design of threaded and welded joints and basic machine elements like shafts, keys, couplings, springs etc.							
4.	Understand, formulate and analyze stresses and strains in machine elements like shafts, keys, couplings, springs etc.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify the need for design process & standards, enumerate different design models.							K3
2.	Design various machine elements against static and fatigue failures							K3
3.	Design bolted and welded joints subjected to various loads.							K3
4.	Evaluate the stresses induced and design the shafts, Keys and couplings.							K4
5.	Design the energy absorbing mechanical components such as springs for the specified loading conditions							K4
SYLLABUS								
UNIT-I	Introduction: Machine Design, basic procedure of machine design, traditional design methods, general design models, BIS system of designation of steels, manufacturing considerations in design. Design against static loads: Modes of failure, Factor of safety, stress strain relationship, shear stress and shear strain, stresses due to bending moment, stresses due to torsional moment, eccentric axial loading, Design of cotter and knuckle joints, Static failure theories							
UNIT-II	Design Against Fluctuating Load: stress concentration, stress concentration factors, reduction of stress concentration, fluctuating stresses, fatigue failure, endurance limit, notch sensitivity, Soderberg, Goodman and modified Goodman diagrams, Gerber equation, fatigue design under combined stresses.							
UNIT-III	Threaded joints: forms of threads, basic types of screw fastenings, ISO metric screw threads, eccentrically loaded bolted joints Welded joints: different types welded joints and their design aspects, Types and strength of weld joints subjected to bending loads and eccentric loads.							
UNIT-IV	Shafts: Shafts design on strength basis, torsional rigidity basis, Design of hollow shafts, flexible shafts, ASME codes for shafts, Keys: keys design, Flat & square keys; Couplings: Rigid and flange couplings, Flexible couplings							

UNIT-V	Spring Design: classification and spring materials, Design of helical compression springs and helical springs subjected to fatigue loading, Design of laminated springs, Equalized stress in spring leaves, Surge in springs, Nipping and shot Peening
Text Books:	
1.	Design of Machine Elements by V.B.Bhandari, TMH Publishing Co. Ltd., New Delhi.
2.	A Text Book of Machine Design by R.S.Khurmi, Eurasia Publishing House Ltd., New Delhi
Reference Books:	
1.	Machine Design by Pandya and Shaw, Charotar publications.
2.	Machine design, an integrated approach by R.L.Norton, Pearson Education
3.	Machine Design by Jain, Khanna Publications.
4.	Mechanical Engineering Design by Joseph Edward Shigley, McGraw-Hill Edition.
Web links	
1.	https://nptel.ac.in/courses/112/105/112105124/
2.	https://nptel.ac.in/courses/112/106/112106137/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3104	PC	3	--	--	3	25	75	3 Hrs.
FLUID MACHINES & SYSTEMS								
(For ME)								
Course Objectives:								
1.	To learn basic concepts of hydraulic machines using impact of jets and jet propulsion							
2.	To study the concepts of different types of turbines, draft tube and performance of turbines							
3.	To study and gain the knowledge on centrifugal pumps, multi stage pumps, priming and cavitation.							
4.	To learn concepts and applications of reciprocating pumps							
5.	To study and understand various types of hydraulic devices							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the concepts of impact jets and jet propulsion and its applications in fluid machinery.							K3
2.	Gain the knowledge and apply the various aspects of impulse and reaction turbines.							K3
3.	Apply the knowledge of centrifugal pumps in various practical situations							K3
4.	Understand and analyze the various aspects of Reciprocating pumps							K4
5.	Understand the description and working of various types of hydraulic devices							K3
SYLLABUS								
UNIT-I	Impact of jet and jet propulsion: Impact of jet on stationary surfaces- Impact of jet on hinged surfaces- moving curved vane with tangential entry of water- Radial flow over the vanes- Jet propulsion of tank and ships							
UNIT-II	Hydraulic Turbines: Classification- Pelton wheel- Reaction turbines- Inward and outward radial flow reaction turbines- Francis turbine- Axial flow reaction turbine- Kaplan turbine- Draft tube- Types- Theory- and efficiency of draft tube. Specific Speed: Determination- Significance- Unit quantities- Unit speed- Unit discharge and unit power- Characteristic curves of hydraulic turbines- Constant head curves- Constant speed curves and Iso-efficiency curves- Governing of turbines.							
UNIT-III	Centrifugal Pumps: Main parts- Efficiency- Minimum speed for starting- Multi-stage centrifugal pumps- Specific speed of a centrifugal pump- Priming of a centrifugal pump- Characteristic curves- Main, Operational and constant efficiency curves- Cavitation- Effects- Cavitation in Hydraulic machines.							
UNIT-IV	Reciprocating Pumps: Main parts- Classification- Velocity and acceleration variation in suction and delivery pipes due to piston acceleration- Effect of variation of velocity on friction in suction and delivery pipes- Effect of acceleration in suction and delivery pipes on indicator diagram- Effect of friction- Maximum speed of reciprocating pump- Airvessels, Simple problems on air vessels.							
UNIT-V	Hydraulic Devices - Hydraulic Press, Hydraulic accumulator- Differential hydraulic accumulator, Hydraulic intensifier, Hydraulic ram, Hydraulic lift, Hydraulic crane, Fluid							

	coupling, Hydraulic torque converter, Air lift pump. Hydraulic Circuits- Definition and representation of directional control valves, Hydraulic circuits for control of Single acting, Double acting hydraulic cylinders, Regenerative circuit, Pump unloading circuit.
Text Books:	
1.	Fluid Mechanics and Hydraulic Machinery, by R.K.Bansal, Laxmi Publications
2.	Fluid Mechanics and Hydraulic Machinery, by R.K.Rajput.
Reference Books:	
1.	Fluid Flow Machines, by N.S.Govinda Rao, Tata McGraw Hill publishing company Ltd.
2.	Hydraulics & Fluid Mechanics, P.N. Modi and S.M. Seth, TEXTBOOKS House, Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3105	PC	3	--	--	3	25	75	3 Hrs.
MECHANICAL MEASUREMENTS AND METROLOGY								
(For ME)								
Course Objectives:								
1.	To impart knowledge on various Metrological equipment's available for measurement.							
2.	To impart knowledge on various methods to measure physical variables.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Illustrate the concept of measurements and measuring techniques.							K3
2.	Understand the principles of limits, fits, tolerances and gauges.							K3
3.	Demonstrate the techniques of linear and angular measurement.							K3
4.	Describe various acceptance tests on a machine tools and techniques of surface finish measurement.							K4
5.	Apply the principles of Optics and fibers in measurement.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Measurement System: Concept of measurements, static performance characteristics, accuracy of measurement and its analysis. Applied Mechanical Measurement: Force (analytical balance, proving rings), Torque (dynamometers), strain (opto-mechanical method), Pressure (Mc-leodgauge), Flow (venturimeter, orificemeter, nozzle) and Temperature (bimetallic thermostat, liquid- in-glass, optical pyrometer)							
UNIT-II (10Hrs)	Basics of Metrology: Introduction to metrology, Need of Inspection, Methods of Measurement, Precision and Accuracy , Factors affecting the accuracy of Measurement, Errors in Measurement System of Limits, Fits, Tolerance and gauging: limits, fits, Tolerances and Interchangeability, Plain limit gauges, Taylor's principle of Gauge Design, Comparators: Twisted strip type, Optical lever, Electric, Electronic, Pneumatic and Optical							
UNIT-III (10 Hrs)	Screw Thread and Gear Metrology: Measurement of screw threads: Major diameter, Minor diameter, Effective diameter and pitch for Internal and External Threads, Measurement of spur gears: pitch, profile, lead, backlash, tooth thickness. Linear and Angular Measurements: Toolmaker's microscope, Slip gauges, Optical bevel protractor, Sine bar, Angle gauges, Precision level, Autocollimator, Angle dekkor, Straightness measurement, Squareness testing, Flatness measurement, Roundness measurement							
UNIT-IV (10 Hrs)	Metrology of Machine Tools: Acceptance tests on Lathe, Milling machine and Radial drilling machine. Measurement of Surface Finish: Surface texture and its Elements, factors effecting the surface roughness, sampling length, Evaluation of surface finish, Principle of Stylus instrument, Stylus probe instruments (Profilometer, Tomlinson surface meter, Taylor Hobson Talysurf)							

UNIT-V (10 Hrs)	Introduction to fiber optics , fiber types, properties of optical fibers and a fiber optic sensor configuration Advances in Metrology : Introduction, Laser beam as a light pointer, length/displacement measurement, temperature sensors, seismographic measurement.
Text Books:	
1.	Mechanical Measurements, by R.S.Sirohi, H.G. Radha Krishna, Wiley Eastern, New Delhi.
2.	A Text Book of Engineering Metrology, I.C.Gupta, Dhanpat Rai & Sons, Delhi.
3.	A Text book of Engineering Metrology, M. Mahajan, Dhanpat Rai & co.
4.	Process & Materials of Manufacture, R.A.Lindberg, 4th edition, Prentice-Hall of India, New Delhi.
Reference Books:	
1.	A.S.T.M.E., Handbook of Industrial Metrology, Prentice-HallofIndia, New Delhi
2.	A.S.T.M.E., Handbook of Manufacturing Engineering.
3.	Manufacturing Processes & Materials for Engineers, L.E.Doyle & others, Prentice-HallofIndia, New Delhi
4.	Manufacturing Technology by Adithan, New age International.
5.	Instrumentation for Engineering Measurements, by R.H.CerniandL.E.Foster, J.Wiley&Sons, New York.
Web links	
1.	https://nptel.ac.in/courses/112/106/112106138/
2.	https://metrology.news/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3106	PC	--	--	3	1.5	20	30	3 Hrs.
IC ENGINES & KINEMATICS OF MACHINES LAB								
(For ME)								
Course Objectives:								
1.	Students would appreciate the fundamentals of thermodynamics being extended to real time applications.							
2.	Students might come out with innovative ideas which may be extended in the form of projects.							
3.	Students by themselves should be able to trouble shoot basic essential issues of automobiles.							
Course Outcomes								
S.No.	Outcome							Knowledge Level
1.	To enhance the knowledge of students by giving hands-on experience with different test rigs of I.C Engines.							K3
2.	To expose the students to different methods of finding friction power in single and multi-cylinder engines.							K5
3.	To estimate the economical speed of different engines.							K5
4.	To evaluate the Gyroscopic couple, static & dynamic balancing methods and four bar chain mechanisms used in various applications.							K5
LIST OF EXPERIMENTS								
(a)	IC Engines Lab							
1.	Load test on single cylinder diesel Engine.							
2.	Morse test on dual cylinder horizontal diesel engine.							
3.	Heat balance sheet on I.C. Engines.							
4.	Study of multi-cylinder engines and determination of its firing order							
5.	Performance test on multi cylinder diesel engine							
6.	To rectify basic essential issues of modern cars such as Jump starting,Tyre replacement & repair, dismantling some components to understand their mechanisms etc.							
(b)	Kinematics of Machines Lab							
1.	Study of automobile mechanisms							
2.	Verification of laws of balancing.							
3.	Determination of ratios of angular speeds of shafts connected by Hooke's joint.							
4.	Determination of the ratio of times and ram velocities of Whitworth quick return motion mechanism.							
5.	To draw curves of slider displacement and crank angle and linear velocities w.r.t. time for a slider crank mechanism and compare with theoretical values.							
6.	To determine the relation of gyroscopic couple and compare with the theoretical values.							
Textbooks:								
1.	Thermal Engineering, by R.K.Rajput.							
2.	Theory of Machines by S S Rattan							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3107	PC	--	--	3	1.5	20	30	3 Hrs.
METROLOGY LAB								
(For ME)								
Course Objectives:								
1.	To demonstrate the fundamentals of Metrology and Measurement Engineering concepts.							
2.	To demonstrate the techniques and processes followed for calibrating of measuring instruments and to familiarize working with different tools used for linear and angular measurement.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Calibrate the accuracy of measuring devices.							K4
2	Calculate linear and angular measurements using different measuring instruments.							K4
SYLLABUS								
1.	Calibration of Vernier Calipers							
2.	Calibration of Outside Micrometer.							
3.	Calibration of Tool Room Microscope.							
4.	Calibration of Mechanical Comparator.							
5.	Measurement of taper using Dial Gauge and Sine Bar.							
6.	Study and use of Bevel Protractor.							
7.	Measurement of the height of Circular Spigot.							
8.	Measurement of angle of V-Groove.							
9.	Measurement of distance between two holes of a template using Vernier Height Gauge							
10.	Measurement of gear parameters							
11.	Establishing thread details on Profile Projector							
12.	Measurement of tool angles on Profile Projector.							
Reference Books:								
1.	A Text Book of Engineering Metrology, I.C.Gupta, Dhanpat Rai & Sons, Delhi.							
2.	Mechanical Measurements, by R.S.Sirohi, H.G. Radha Krishna, Wiley Eastern, New Delhi							
3	Manufacturing Processes & Materials for Engineers, L.E.Doyle & others, Prentice-Hall of India, New Delhi							
4	Instrumentation for Engineering Measurements, by R.H. Cerni and L.E.Foster, J.Wiley & Sons, New York.							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS I								
Common to All Branches								
Part-A: Verbal and Soft Skills-I								
Course Objectives:								
1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.							
2.	To familiarize the learner with high frequency words as they would be used in their professional career.							
3.	To inculcate logical thinking in order to frame and use data as per the requirement							
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.							
5.	To familiarize students with soft skills and how it influences their professional growth.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.							K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.							K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.							K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.							K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.							K3
SYLLABUS								
UNIT-I	Grammar: (VA) Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns- number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.							
UNIT-II	Vocabulary: (VA) Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.							

UNIT-III	Reasoning: (VA) Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).
UNIT-IV	Usage: (VA) Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	Soft Skills: Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners,,s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal,,s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.
Part-B: Quantitative Aptitude-I	
Course Objectives:	
1.	To familiarize students with basic problems on numbers and ratios problems.
2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4.	To inculcate logical thinking by exposing the students to reasoning related questions.
5.	To inculcate logical thinking by exposing the students to reasoning related questions.

Course Outcomes:		
S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	
UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.	
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.	

Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

MECHANICAL ENGINEERING
(Accredited by NBA)
SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)
III/IV B.TECH
II-SEMESTER
(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr.	L	T	P	Internal Marks	External Marks	Total Marks
B19 ME 3201	Industrial Engineering & Management	PC	3	3	--	--	25	75	100
B19 ME 3202	Dynamics of Machines	PC	3	3	--	--	25	75	100
B19 ME 3203	Machine Design	PC	3	3	--	--	25	75	100
# PE-I	Program Elective-I	PE	3	3	--	--	25	75	100
#PE-II	Program Elective-II	PE	3	3	--	--	25	75	100
#OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 ME 3212	Industrial Engineering Lab	PC	1.5	--	--	3	20	30	50
B19 ME 3213	Fluid Mechanics and Machinery Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3201	Employability Skills-II	MC	--	3	--	--	--	--	--
B19 MC 3202	Basic Coding	MC	--	--	--	3	--	--	--
TOTAL			21	21	--	9	190	510	700

	Course Code	Course
#PE-I	B19 ME 3204	Composite Materials
	B19 ME 3205	Finite Element Analysis
	B19 ME 3206	Production Planning and Control
	B19 ME 3207	MOOCs –I
#PE-II	B19 ME 3208	Automobile Engineering
	B19 ME 3209	Tool Design
	B19 ME 3210	Unconventional Machining Processes
	B19 ME 3211	MOOCs –II
#OE-I	Student has to study one Open Elective offered by CE or CSE or ECE or EEE or IT or EM&H from the list enclosed.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3201	PC	3	--	--	3	25	75	3 Hrs.
INDUSTRIAL ENGINEERING AND MANAGEMENT								
(For ME)								
Course Objectives: Students learn								
1.	To know the management practices and to handle the industrial disputes in industry							
2.	To know the production activities and maintenance practices in industry							
3.	To know the work study procedures and quality concepts to increase productivity							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the knowledge of industrial and personal management for the settlement of industrial disputes							K3
2.	Develop the production planning activities for the advancement of productivity							K3
3.	Illustrate various economics of plant layout for the better maintenance of plant							K3
4.	Interpret the concepts of work-study to evaluate the performance							K4
5.	Apply the tools and techniques of quality management to manufacturing and quality control							K4
SYLLABUS								
UNIT-I (15 Hrs)	Concepts of Industrial Management: Principles of Management, Functions of management, Principles of organization, Types of organization Personnel management and Industrial relations: Functions of personnel management, Theories of motivation, Promotion, Transfer, Lay off and Discharge, Labor turnover, Trade unions, Industrial disputes, Strikes, Lock-out, Picketing, Gherao, Settlement of industrial disputes, Collective bargaining, Industrial dispute act 1947 and Factories act 1948.							
UNIT-II (08 Hrs)	Production planning and control: Types of productions, Production cycle, Product design and development, Process planning, Forecasting, Loading, Scheduling, Dispatching, Routing, Progress control.							
UNIT-III (10 Hrs)	Plant location & Plant layout: Economics of plant location, Rural Vs Suburban sites, Types of layouts, Types of buildings. Materials handling: Principles, Concept of unit load, Palletization and Containerization, Selection of material handling equipment, Types and applications of material handling equipment.							
UNIT-IV (10 Hrs)	Work study: Concept of productivity, Method study – Basic steps in method study, Process charts, Diagrams, Models and Templates, Principles of motion economy, Micro motion study, Therbligs, SIMO chart, Work measurement – Stop watch procedure of time study, Performance rating, Allowances, Work Sampling-Simple problems.							
UNIT-V (10 Hrs)	Materials management: Introduction, Purchasing, Objectives of purchasing department, Buying techniques, Purchase procedure, Stores and material control, receipt and issue of materials, Store records, introduction to MRP and MRP-II. Quality control: Control charts of variables and attributes. Single and double sampling plans.							

Text Book:	
1.	Industrial Engineering and Management by Dr. O P Khanna
Reference Books:	
1.	Principles of Management by Koontz &Donnel.
2.	Production and Operations Management by Everette Adam & Ronald Ebert
3.	Industrial Engineering and Production Management by MartandTelsang
4.	Operations Management by John McClain & Joseph Thames.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3202	PC	3	--	--	3	25	75	3 Hrs.
DYNAMICS OF MACHINES								
(For ME)								
Course Objectives:								
1.	To understanding the effect of gyroscopic couple on stabilization of sea vehicles, aircrafts and automobile vehicles							
2.	To develop the knowledge of analytical and graphical methods for calculating balancing of rotary and reciprocating masses.							
3.	To understand the concept of vibrations, bearings, clutches and effect of friction in motion transmission and its significance on engineering design.							
4.	To understand the concept of turning moment diagrams, flywheel and speed control systems such as governors							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the principles of gyroscopic couple on stabilization of sea vehicles, aircrafts and automobile vehicles							K3
2.	Analyze the principles of balancing to rotating and reciprocating members							K4
3.	Illustrate the vibrations of single degree of freedom of mechanical systems &Friction to Clutches, mechanical components and Bearings							K3
4.	Analyze the basic principles of mechanisms to Governors, flywheel & forces on engine components and Turning moment diagrams							K4
SYLLABUS								
UNIT-I (10 Hrs)	GYROSCOPIC COUPLE AND PRECESSIONAL MOTION: Precessional and angular motion- gyroscopic couple- effect of gyroscopic couple on an aero plane and on a naval ship, stability of a four-wheel vehicle moving in a curved path, stability of a two-wheel vehicle taking a turn.							
UNIT-II (10 Hrs)	BALANCING OF ROTATING AND RECIPROCATING MASSES: Balancing of a single rotating mass in the same plane and by two masses in different planes, balancing of several masses revolving in the same plane- Balancing of several masses revolving in different planes- Primary and secondary unbalanced forces of reciprocating masses, Partial balancing of unbalanced primary forces in a reciprocating engine, Partial balancing of locomotives- Effect of partial balancing of reciprocating parts of two cylinder locomotives- Variation of tractive force, Swaying couple and hammer blow- Balancing of primary and secondary forces in multi cylinder in-line engines- Direct and reverse cranks- Balancing of V- Engines							
UNIT-III (10 Hrs)	VIBRATIONS: Definitions- Types of vibrations- Natural frequencies of free longitudinal vibrations of systems having single degree of freedom- Equilibrium method- Energy method and Rayleghy's method. Frequency of damped vibration and forced vibration with damping- , Simple problems on forced damped vibration, Magnification factor . Natural frequency of free transverse vibrations due to point load and uniformly distributed load acting over a simply supported shaft- Transverse vibrations for a shaft subjected to number of point loads- Energy method- Dunkerley's method, Critical speed of a shaft. Natural frequency of free torsional vibrations- Free torsional vibrations of single rotor system							

UNIT-IV (10 Hrs)	FRICTION: Introduction; Kinds of friction; Laws of friction; Coefficient of friction; Screw jack; friction of screw, Pivots and collars; Rolling friction; Antifriction bearings; Greasy friction; Greasy friction at a journal; Film friction; CLUTCHES: Friction clutches- single disc or plate clutch, multiple disc clutch, cone clutch, centrifugal clutch.
UNIT-V (10 Hrs)	TURNING MOMENT DIAGRAMS: Analytical method for velocity and acceleration of piston, angular velocity and acceleration of connecting rod, Engine force analysis, crank effort and turning moment diagrams – fluctuation of energy – fly wheels. GOVERNERS: Watt, porter and proell governors, spring loaded governors – Hartnell sensitiveness, isochronism and hunting Stability;
Text Books:	
1.	Theory of Machines / S.S Ratan/ Mc. Graw Hill Publ
2.	Theory of machines / Khurmi / S.Chand.
3.	Mechanism and machine theory by Ashok G. Ambedkar, PHI Publications
Reference Books:	
1.	Mechanism and Machine Theory / JS Rao and RV Dukupati / New Age.
2.	Theory of Machines / Shiegly / MGH
3.	Theory of Machines / Thomas Bevan / CBS Publishers

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3203	PC	3	--	--	3	25	75	3 Hrs.
MACHINE DESIGN								
(For ME)								
Course Objectives:								
1.	To Design and formulate, to analyze stresses and strains in machine elements like gears, clutches, brakes, bearings, etc. under static and dynamic load conditions.							
2.	To Design of Internal combustion engine parts like cylinder, pistons, connecting rod etc							
3.	To understand the stresses on miscellaneous machine elements like wire ropes, chain drives etc.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Design different types of gears such as spur, helical & bevel gears on the basis of their load carrying capacity.							K4
2.	Design various parts of IC Engines such as cylinders, pistons, connecting rods & cross heads.							K4
3.	Apply the design concepts to determine the various parameters of clutches.							K4
4.	Apply the design concepts to determine the torque and dimensions related to brakes.							K4
5.	Design the sliding and roller contact bearings under various environmental and service Conditions.							K4
6.	Select and Design wire ropes and chain drives for the specified application.							K4
SYLLABUS								
UNIT-I	Gears: Classification of gears, Standard tooth systems. Spur, Helical and Bevel gears. Terminology of each. Tooth failure. Face width and beam strength. Lewis equation. Design for dynamic and wear loads. Force analysis of Bevel gears							
UNIT-II	I.C Engine elements: Design of cylinders and heads, Design of pistons, Design of cross-head and Design of connecting rods							
UNIT-III	Clutches: Torque capacity of single and multi-plate clutches. Design considerations. Energy considerations and Temperature rise friction materials. Cone clutch design, Centrifugal clutches design. Brakes: Energy equations, self locking and self energizing conditions of a brake. Design of Single & Double block brakes, Simple & Differential band brakes and Internal expanding shoe brakes.							
UNIT-IV	Sliding contact bearings: Lubrication modes. Temperature effect on viscosity. Journal bearing design. Bearing modulus. McKee equations. Heating of bearings. Collar and thrust bearings. Rolling Contact Bearings: Static and dynamic load capacity. Equivalent bearing load. Load-life relationships. Load factor. Selection of bearings from manufacturer's catalogue.							
UNIT-V	Wire ropes: construction, Classification and designation, Stresses in wire ropes, Design for service, advantage of wire ropes over fiber ropes Chain drives: Terminology and classification, Design procedure for service. Advantages and disadvantages of chain drive over rope drives							

Text Books:	
1.	Design of Machine Elements by V.B.Bhandari, TMH Publishing Co. Ltd., New Delhi.
2.	A Text Book of Machine Design by R.S.Khurmi, Eurasia Publishing House Ltd., New Delhi
Reference Books:	
1.	Machine Design by Pandya and Shaw, Charotar publications.
2.	Machine Design by Jain, Khanna Publications.
3.	Machine design, an integrated approach by R.L.Norton, Pearson Education.
4.	Mechanical Engineering Design by Joseph Edward Shigley, McGraw-Hill Edition.
Web links	
1.	https://nptel.ac.in/courses/112/105/112105124/
2.	https://nptel.ac.in/courses/112/106/112106137/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3204	PE	3	--	--	3	25	75	3 Hrs.
COMPOSITE MATERIALS								
(Program Elective-I)								
(For ME)								
Course Objectives:								
1.	To impart knowledge about the composite materials and their properties and structure.							
2.	To develop the student's skills in understanding the different manufacturing methods available for Metal Matrix and advanced composites.							
3.	Illuminate the knowledge and analysis skills to understand the response of composites to stresses.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify the functional requirements of reinforcement and matrix and classification of composites based on structure and matrix.							K3
2.	Identify and describe the properties of fibers and manufacturing of polymer matrix composites.							K3
3.	Develop competency in one or more common composite manufacturing techniques, and be able to select the appropriate technique for manufacturing of metal matrix composites.							K3
4.	Determine stresses and strains relation in composites materials.							K3
SYLLABUS								
UNIT-I (10 Hrs)	INTRODUCTION: Definition, Classification of Composite materials based on structure and based on matrix. Advantages of composites, application of composites, functional requirements of reinforcement and matrix.							
UNIT-II (10 Hrs)	FIBERS: Preparation, properties and applications of glass fibers, carbon fibers, Kevlar fibers and metal fibers, properties and applications of whiskers, particle reinforcements. Introduction to natural fibers.							
UNIT-III (10 Hrs)	MANUFACTURING OF ADVANCED COMPOSITES: Polymer matrix composites: Preparation of Moulding compounds and pre-pegs, hand layup method, Autoclave method. Filament winding method, Compression moulding, Reaction injection moulding.							
UNIT-IV (10 Hrs)	MANUFACTURING OF METAL MATRIX COMPOSITES: Casting, Solid State diffusion technique, Cladding - Hot isostatic pressing. Manufacturing of Ceramic Matrix Composites: Liquid Metal Infiltration, Liquid phase sintering. Manufacturing of Carbon – Carbon composites: Knitting, Braiding, Weaving.							
UNIT-V (10 Hrs)	RESPONSE OF COMPOSITES TO STRESS: (a) Iso Strain condition (b) Iso Stress condition (c) Load friction shared by the fibers. MICRO MECHANICAL ANALYSIS OF A LAMINA: Introduction, Evaluation of the four elastic moduli by Rule of mixture, Numerical problems. Macro Mechanics of a Lamina: Hooke's law for different types of materials, Number of elastic constants, Two - dimensional relationship of compliance and stiffness matrix.							

Text Books:	
1.	Material Science and Technology, Vol. 13, 3rd edition- V. C. H. Cahn (2007).
2.	Composite Materials, 2nd edition, Springer, USA - K. K. Chawla (2010).
Reference Books:	
1.	Hand Book of Composite Materials, 3rd edition, Tata McGraw-Hill - E. D. Lubin (2003).
2.	Composite Materials: Processing and Technology, 2nd edition, Academy Press - Muhammad M. Rafique (2009).
3.	Composite Materials and structure, IIT Kharagpur, India - P. K. Sinha (2006).
Web links	
1.	https://nptel.ac.in/courses/112/104/112104229/
2.	https://nptel.ac.in/courses/112/104/112104168/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3205	PE	3	--	--	3	25	75	3 Hrs.
FINITE ELEMENT ANALYSIS								
(Program Elective-I)								
(For ME)								
Course Objectives:								
1.	To provide students with a conceptual understanding of the principles of finite element analysis systems, the implementation of these principles, and its connections to CAD.							
2.	To teach students how to perform structural analysis using finite element methods.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand the fundamental concepts of Finite Element Analysis and Solve the physical problem using functional approximation method.							K3
2.	Analyze the 1Dstructural problems by applying the concepts of finite element analysis.							K4
3.	Analyze Trusses and Beams by applying the concepts of finite element analysis.							K4
4.	Analyze 2D structural problems by applying concepts of finite element analysis and apply the principles of Numerical Integration and its application to Finite Element Analysis							K4
5.	Analyze Axisymmetric solids by applying the concepts of Finite Element Analysis.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: stress and equilibrium, strain – displacement relations, stress – strain relations, plane stress and plane strain conditions, The potential energy approach; Rayleigh-Ritz method, Galerkin’s method, Weighted Residual method, Finite Element Method: Discretization, Types of elements, band width, node numbering, interpolation functions, local and global coordinates, convergence requirements, Types of boundary conditions, Steps in Finite Element Method, Applications of Finite Element Method .							
UNIT-II (10 Hrs)	One Dimensional Bar Problems: 1-D bar element - shape functions – Stiffness matrix and load vector– assembly of Matrices – Treatment of boundary conditions One dimensional quadratic element – Temperature Effects.							
UNIT-III (10 Hrs)	Trusses: Introduction; Plane trusses; shape functions – Stiffness matrix and load vector– assembly of Matrices – Treatment of boundary conditions; simple problems on trusses. Analysis of Beams: Beam Element - Shape functions and Element stiffness matrix, load vector for concentrated and Uniformly Distributed Load, simple problems on beams.							
UNIT-IV (10 Hrs)	Two Dimensional Problems: Finite element modeling of two-dimensional Problems - constant strain triangle Element - treatment of boundary conditions 2D four noded iso parametric element, numerical integration, Gaussian Quadrature Approach.							
UNIT-V (10 Hrs)	Axisymmetric Solids Subjected to Axisymmetric Loading: Introduction; Axisymmetric formulation; Finite element modeling - triangular element; Problem modeling and boundary conditions.							

Text Books:	
1.	Introduction to Finite Elements in Engineering, Chandrupatla, Ashok and Belegundu, Prentice – Hall.
Reference Books:	
1.	The Finite Element Method by O.C. Zienkiewicz, Tata McGraw Hill Company Ltd.
2.	The Finite Element Methods in Engineering by Rao, S.S.
3.	Concepts and Applications of Finite Element Analysis by Cook, R.D.
4.	Applied Finite Element Analysis by Segerland, L.J.
Web links	
1.	https://nptel.ac.in/courses/112/104/112104116/
2.	https://nptel.ac.in/courses/112/104/112104193/
3.	https://nptel.ac.in/courses/112/104/112104205/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3206	PE	3	--	--	3	25	75	3 Hrs.
PRODUCTION PLANNING AND CONTROL								
(Program Elective-I)								
(For ME)								
Course Objectives:								
1.	To develop an ability to apply PPC concepts in a various area like marketing, accounting, finance, engineering, personnel management, logistics, etc.							
2.	To examine several classic Operations Management planning topics including production planning and inventory control.							
3.	To aquire knowledge to make MRP, MRP-II by using Modern Production planning software for production of product with available resources.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Judge which type of production is required for different specific real-world situations and can apply various qualitative/quantitative methods for forecasting the future demand.							K3
2.	Analyze different inventory systems for minimizing the total costs and maximizing the profit.							K3
3.	Determine the most economical process of doing a work and prepare the route sheets for establishing how and where the work will be done.							K3
4.	Analyze the sequence of performing jobs scheduled through different machines to measure the effectiveness of the system and can also administer the priority rules for dispatching jobs.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Definition – Objectives of production Planning and Control – Functions of production planning and control – Types of production – Organization of production planning and control department.							
UNIT-II (10 Hrs)	Forecasting: Importance – Types of forecasting– Forecasting techniques – qualitative methods and quantitative methods, Forecasting errors							
UNIT-III (10 Hrs)	Inventory management: Functions of inventories – relevant inventory costs – EOQ model – Inventory control systems: Fixer order quantity system and Periodic review system - ABC analysis -VED analysis- Material Requirement Planning, Bill of material, MRP II - Master Production Scheduling.							
UNIT-IV (10 Hrs)	Aggregate planning: Chase planning, Expediting, controlling aspects. Routing: Definition – Routing procedure –Route sheets — Factors affecting routing, procedure – Difference with loading							
UNIT-V (10 Hrs)	Scheduling: Policies – Types of scheduling - Forward and Backward Scheduling – Gantt Charts – Flow shop Scheduling – n jobs and 2 machines, n jobs and 3 machines – Job shop Scheduling – 2 jobs and n machines – Line of Balance. Dispatching: Activities of dispatcher – Dispatching procedure – follow up – priority rules for dispatching jobs - Applications of computer in production planning and control.							

Text Books:	
1.	Elements of Production Planning and Control / Samuel Eilon.
2.	Modern Production/ operation managements / Baffa & Rakesh Sarin
3.	Production Planning & Control – Jain and Agarwal
Reference Books:	
1.	Operations Management – S.N. Chary. 2. 3. 4.
2.	Inventory Control Theory and Practice / Martin K. Starr and David W. Miller.
3.	Production Control A Quantitative Approach / John E. Biegel.
4.	Operations Management / Joseph Monks.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3207	PE	--	--	--	3	25	75	3 Hrs.
MOOCS-I								
(For ME)								
MOOCs-I course should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3208	PE	3	--	--	3	25	75	3 Hrs.
AUTOMOBILE ENGINEERING								
(Program Elective-II)								
(For ME)								
Course Objectives:								
1.	To make students familiar with the constructional details of chassis and body.							
2.	To understand about various steering systems, steering linkages, Transmission system, steering gear boxes and power steering.							
3.	To study the different components in the drive line and types of final drive.							
4.	To introduce students to the rear axles and types of suspension systems.							
5.	To introduce students to braking systems, wheels and tyres and provides the information on various aspects of vehicle maintenance.							
6.	To understand upcoming technology of hybrid electric vehicles.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Identify the Automobile layout, types of IC engines and their subsystems.							K3
2.	Analyze various types and working principles of clutch, gearbox, drive shaft and final drive systems.							K3
3.	Illustrate key elements of Steering geometry in automobile and can compare various types of steering mechanism, suspension systems, wheels, tires based on their construction.							K3
4.	Analyze the concepts of brakes, electrical and electronic systems, pollution control methods and norms.							K3
5.	Identify technological updates in Hybrid Vehicles and the need of trouble shoot and maintenance in Automotive Vehicles.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Automobile, Automobile Layout, Chassis and body, Power unit- types of automobile engines, engine parts, Classification: ‘In-line’ and ‘V’ type, Multi-Valve Engines, Super Charging/Turbo charging, Air filters, Fuel Systems: Petrol Engines: Carburetted and MPFI, Diesel Engines: Conventional, CRDI and Dual fuel Engines, Engine Cooling and Lubrication.							
UNIT-II (10 Hrs)	Clutches: principle, Types: cone clutch, single plate clutch, diaphragm clutch, multi plate clutch, centrifugal clutches and fluid coupling. Gearbox: Construction and Working Principle, Selector Mechanism, Types: Sliding mesh, Constant mesh, Synchromesh, and Epicyclical, Overdrive, Automatic Gearbox-CVT, Torque converter. Drive shaft and Final Drive: Drive Shaft, Types of Propeller shafts, Final drive and Differential, Power transmission: Front, Rear and Four wheel drive.							
UNIT-III (10 Hrs)	Suspension System: Leaf springs coil springs, torsion bar, shock absorber, Independent suspension system.							

	Steering System: Steering geometry: camber, caster, Kingpin angle, Toe-in, and Toe-out. Steering Mechanism and its Elements: Steering gear box and its types, Steering gear ratio, Power-Steering Wheels: Disc and Drum type, Tires: Tire Construction, Tube and Tubeless Tires, Radial Tires, Tire specification, Tire rotation and Tire Maintenance.
UNIT-IV (10 Hrs)	Braking System: Necessity, Parking and Power Brakes, Parts and Working Principle of Mechanical, Air and. Hydraulic Brakes: Master and Wheel cylinder, Properties of Brake Fluids, Brake Diagnostics and Service: Brake Bleeding, Anti-lock Braking System. Air pollution and their control: EGR and Catalytic Converters, EURO/Bharat Stage Norms, Mufflers. Electrical and Electronic system: Starting System, Ignition system, battery, ECU/ ECM.
UNIT-V (10 Hrs)	Hybrid Vehicles: History and Introduction of Hybrid Vehicles, Components in hybrid vehicles, Classification of hybrid topologies- Drivetrain structure, Degree of hybridization, Nature of the power source, Advantages and Disadvantages, Applications. Trouble shooting and Maintenance: Engine and Vehicle Troubles: Diagnostic Information, Symptom descriptions and their Causes and Remedies, Maintenance - Periodic, Preventive and Break down.
Text Books:	
1.	Automotive Mechanics (10/e) - William H. Crouse and Donald L. Anglin, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-059054-0.
2.	Automobile Engineering – KK Jain/ RB Asthana, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-044529-X.
3.	Internal Combustion Engines and Air Pollution- E.F. Obert, Harper & Row International Publishers Inc., ISBN: 0-06-350561-4.
4.	Electric and Hybrid Vehicles, Tom Denton, Taylor & Francis, 2018.
Reference Books:	
1.	Automotive Mechanics – S. Srinivasan, Tata McGraw-Hill Publishing company Limited, ISBN: 0-07-044941-6.
2.	Internal Combustion Engines – Heywood, John, B. McGraw-Hill Publications Limited.
3.	Automotive Engines- S Srinivasan, Tata McGraw-Hill Publishing Company Limited, ISBN: 0-07-040265-5.
4.	Hybrid Vehicles and the future of personal transportation, Allen Fuhs, CRC Press, 2011.
Web links	
1.	https://nptel.ac.in/courses/107/106/107106088/
2.	https://nptel.ac.in/courses/108/103/108103009/
3.	https://www.theengineerspost.com/category/automobile-engg/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3209	PE	3	--	--	3	25	75	3 Hrs.
TOOL DESIGN								
(Program Elective-II)								
(For ME)								
Course Objectives:								
1.	To apply the principles of locating and clamping systems.							
2.	Apply the design of different tools and components, such as jigs and fixtures, press tools for sheet metal working, molds for plastic injection molding, and die casting.							
3.	Use of limit gauges and tolerances							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain about locating and clamping devices.							K3
2.	Practice with jigs and fixtures.							K3
3.	Use press and press tools and design different types of dies							K3
4.	Illustrate Die casting Dies and Injection Moulds							K3
5.	Determine gauges and guage design.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Locating and Clamping Devices: Principles of Jigs and Fixtures design-Locating principles-Locating elements-Standard parts-Clamping devices-Mechanical actuation-Pneumatic & hydraulic actuation-Analysis of clamping forces-Tolerance and error analysis.							
UNIT-II (10 Hrs)	Jigs & Fixtures: Drill bushes-Different types of Jigs-Plate latch, channel, box, post, angle plate, angular post, turnover, pot jigs- Automatic drill jigs-Rack & Pinion Operated, Air operated Jigs Components. General principles of lathe, milling and broaching fixtures-Grinding, Drilling and shaping fixtures, Assembly, Inspection and Welding fixtures-Modular fixtures. Design and development of Jigs and fixtures for simple components.							
UNIT-III (10 Hrs)	Press Tools: Press working terminology-Presses and Press accessories-Computation of capacities and tonnage requirements-Design and development of various types of cutting, forming and drawing dies. Bending dies – Introduction, bend allowance, spring back, edge bending die design. Drawing dies – Single action, double action and triple action dies, factors affecting drawing, drawing die design.							
UNIT-IV (10 Hrs)	Die Casting Dies: Basic Terminology, Types of Dies: Single cavity, multicavity dies, combination dies, unit dies, Pressure Die casting Process, Requirements, Technique of filling Die Cavity, Pressure Die casting machines, advantages and disadvantages of Pressure die Casting Process, defects in die casting. Injection Moulding: Basic Terminology, Injection moulding machine and its elements, general configuration of a mould. 2 plate and 3 plate mould. Introduction to compression, transfer, blow moulding, extrusion, forming and calendaring.							
UNIT-V (10 Hrs)	Design of Limit Gauges: Elements, types and application of limit gauges, Gauge materials, their selection, Taylor’s principles of gauge design, Types and methods to provide gauge tolerances. Design steps and design of plug & ring / snap gauge for given dimension and application.							

Text Books:	
1.	Donaldson. C, Tool Design, Tata McGraw-Hill, 1986
2.	“ASTME Handbook of Fixture Design ".Prentice Hall of India Pvt. Ltd.
3.	Basu, Mukherjee, Mishra, Fundamentals of Tool Engg. Design, Oxford & IBH Publishing, N. Delhi
Reference Books:	
1.	A. K. Goroshkin, " Jigs and Fixtures Handbook ", Mir Publishers, Moscow, 1983.
2.	“Die Design Handbook ", Ivana Suchy, McGraw Hill Book Co., 2005.
3.	Injection Moulding Design, RGW Pye, john.1998
4.	Production technology, HMT,Tata McGraw Hill.
5.	P. Eugene Ostergaard, “Basic Die Making” - Mc Graw Hill Book, 1963.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3210	PE	3	--	--	3	25	75	3 Hrs.
UNCONVENTIONAL MACHINING PROCESSES								
(Program Elective-II)								
(For ME)								
Course Objectives:								
1.	To impart the student the knowledge of unconventional machining methods and their applications.							
2.	To acquaint the student with the knowledge of different Mechanical energy based unconventional machining methods.							
3.	To acquaint the students with the concepts of Thermal and Thermo-electrical energy based unconventional machining process.							
4.	To impart the student the concept of different Electro-Chemical and Chemical energy based unconventional machining process.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Distinguish the types of unconventional machining processes and their applications.							K3
2.	Describe the unconventional machining process using Mechanical energy and its applications.							K3
3.	Distinguish the process of unconventional machining using Thermal energy and Thermo-electrical energy and their applications.							K3
4.	Demonstrate the process of unconventional machining using electro-chemical and chemical energy and their applications.							K3
SYLLABUS								
UNIT-I (06 Hrs)	Introduction–Need for non-traditional machining methods – Classification of modern machining processes– considerations in process selection-Materials –Applications.							
UNIT-II (12 Hrs)	Abrasive Jet Machining, Water Jet Machining and Abrasive Water Jet Machine: Basic principles- equipment-process variables- and mechanics of metal removal- MRR-application and limitations. Ultrasonic machining– Elements of the process – mechanics of metal removal–process parameters – economic considerations – applications and limitations – recent developments.							
UNIT-III (10 Hrs)	Electric Discharge Machining–Power Circuits – Mechanics of metal removal – Process parameters – selection of tool electrode and dielectric fluids – flushing – applications – Wire EDM – principle – applications.							
UNIT-IV (12 Hrs)	Electron Beam Machining (EBM) -Laser Beam Machining (LBM) –Plasma Arc Machining (PAM) - Principles – Equipment – Process Parameters – Applications. Magnetic abrasive finishing – Abrasive flow finishing – Electro stream drilling – shaped tube electrolyte machining.							
UNIT-V (10 Hrs)	Electro–Chemical Machining Processes: Principles of ECM –equipment – MRR – process parameter –electrochemical grinding – electro chemical honing and deburring process.							

	Chemical machining –principle–etchants –maskants– maskant application methods – process parameter – MRR – applications.
Textbooks:	
1.	Advanced Machining Processes/VKJain/Alliedpublishers
2.	Modern Machining Processes-P. C.Pandey, H. S.Shan
Reference Books:	
1.	Manufacturing Engineering and Technology By Serope Kalpak Jain,PearsonPublications.2001
2.	Manufacturing Engineering & Technology, Kalpak Jain
3.	Unconventional Manufacturing Processes, Singh M. K

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3211	PE	--	--	--	3	25	75	3 Hrs.
MOOCS-II								
(For ME)								
MOOCs-II course should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	OE	3	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE-I								
(For ME)								
Student has to study one Open Elective offered by CE or CSE or ECE or EEE or IT or EM&H from the list enclosed.								

Offered by	Course Code	Course Name	Offered to
CIVIL ENGINEERING	B19CEOE01	Disaster Management	CSE,ECE,EEE, IT &ME
	B19CEOE02	Remote Sensing &GIS	
COMPUTER SCIENCE & ENGINEERING	B19CSOE01	Data Structures	CE,ECE,EEE, &ME
	B19CSOE02	Java Programming	
	B19CSOE02	Web Technologies	
ELECTRONICS & COMMUNICATION ENGINEERING	B19ECOE01	Basic Electronics	CE,CSE,EEE, IT &ME
	B19ECOE02	Signals & Systems	
ELECTRICAL & ELECTRONICS ENGINEERING	B19EEOE01	Introduction To Electrical Systems	CE,CSE,ECE, IT &ME
	B19EEOE02	Electrical Estimation and Costing	
INFORMATION TECHNOLOGY	B19ITOE01	Data Structures & Algorithms	CE,ECE,EEE, &ME
	B19ITOE02	Database Design & Development	
	B19ITOE03	Java Programming	
ENGINEERING MATHEMATICS & HUMANITIES	B19BSOE01	computational statistics with R	ALL BRANCHES
	B19BSOE02	Fuzzy sets and Fuzzy Logic	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3212	PC	--	--	3	1.5	20	30	3 Hrs.
INDUSTRIAL ENGINEERING LAB								
(For ME)								
Course Objectives:								
1.	To make students acquainted with the different control charts and probability distribution curves							
2.	To make students aware with the different types of process charts for improving method of doing work							
3.	To make students learn the impact of work on human physiology and physiological constraints of body							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Analyze Shewart Normal Bowl theory.							K4
2	Evaluate the process and product quality by Statistical Quality Control Techniques.							K5
3	Practice Work Study Techniques for man, machine and work place to improve productivity.							K3
SYLLABUS								
1.	To show that the sample means from a normal universe follow a normal distribution.							
2.	To show that the sample means from a non-normal universe follow a normal distribution.							
3.	To draw the control charts for mean and range for the measurements of output of a manufacturing process, and to study its process capability.							
4.	To draw the control chart for fraction defective for a given lot of marble balls.							
5.	To draw the control chart for defects observed on a given lot of steel discs.							
6.	To conduct Single Sampling Plan on a given lot of marbles, and hence to draw its operating characteristic curve.							
7.	To draw two handed process charts for (i)Bolt and Washers assembly; (ii) Assembly of electric tester							
8.	To draw Multiple Activity chart using an Electric toaster.							
9.	To draw flow process charts on activities in Workshop/Laboratory/Office.							
10.	To measure the skill and dexterity in the movement of Wrist and Fingers using pin board, and hence to estimate the standard time for the best method of performance.							
11.	To measure the Heart beat using Stethoscope and to study the changes in heart rate for different subjects using Treadmill.							
12.	To measure the Heart rate during working and recovery periods of the subjects under different loads, using Bicycle Ergometer.							
Reference Books:								
1.	Industrial Engineering Management by Dr. O.P. Khanna							
2.	Industrial Engineering and Production Management by Telsay, S.Chand & Co							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19ME3213	PC	--	--	3	1.5	20	30	3 Hrs.
FLUID MECHANICS AND MACHINERY LAB								
(For ME)								
Course Objectives:								
1.	This course will provide a basic understanding of flow measurements using various types of flow measuring devices, calibration and losses associated with these devices							
2.	Energy conversion principles, analysis and understanding of hydraulic turbines and pumps will be discussed. Application of these concepts for these machines will be demonstrated. Performance analysis will be carried out using characteristic curves							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Perform Experiments to determine the coefficient of discharge of flow measuring devices							K3
2	Conduct experiments on hydraulic turbines and pumps to draw characteristic curves							K3
3	Test basic performance parameters of hydraulic turbines and pumps and use the acquired knowledge to exhibit competency towards preventive maintenance of the hydraulic equipment							K4
SYLLABUS								
1.	Flow through (i) Orificemeter and (ii) Venturimeter							
2.	Flow through (i) V Notch and (ii) Rectangular Notch							
3.	Bernoullie’s Theorem							
4.	Major and Minor losses in pipes							
5.	Performance test on reciprocating pump							
6.	Performance test on single stage centrifugal pump							
7.	Impact of jet on vanes							
8.	Performance test on pelton wheel							
Reference Books:								
1.	Fluid Mechanics and Hydraulic Machines by Dr. R.K.Bansal							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS II								
Common to All Branches								
Part-A: Verbal and Soft Skills-II								
Course Objectives:								
1.	To expose the students to bettering sentence expressions and also forming equivalents.							
2.	To instill reading and analyzing techniques for better comprehension of written discourses							
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.							
4.	To inculcate conversational skills, nuances required when interacting in different situations							
5.	To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.							K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences							K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).							K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.							K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.							K3
SYLLABUS								
UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.							
UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author,,s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.							

UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.
Reading/ Listening material:	
1.	Guide to IELTS, Cambridge University Press.
2.	Barron,,s GRE guide
3.	Newspapers like The Hindu, Times of India, Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN
Text Books:	
1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.
Reference Books:	
1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie
5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.
Part-B: Quantitative Aptitude-II	
Course Objectives: The objective of introducing quantitative aptitude –II is:	
1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3.	To familiarize the students with finding out accurate date and time related problems
4.	To enable the students solve the puzzles using logical thinking.
5.	To expose the students to various problems based on geometry and mensuration.

Course Outcomes:		
S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3
SYLLABUS		
UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.	
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.	
UNIT-III	Periods, Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.	
UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles	
UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.	
Text Books:		
1.	Quantitative aptitude by RS Agarwal	
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi	
3.	Puzzles to puzzle you by shakunataladevi	
4.	More puzzles by shakunataladevi	
5.	Puzzles by George summers	

Reference Books:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3202	MC	--	--	3	--	--	--	--
BASIC CODING								
(Common for CE and ME)								
Course Objectives:								
1.	To develop programming skills among the students							
2.	To familiarize the student with Control Structures, Loop Structures							
3.	To familiarize the student with Basic searching and sorting Methods							
4.	To familiarize the student with Functions, Recursions and Storage Classes							
5.	To familiarize the student with Structures and Unions							
6.	. To familiarize the student with Operating System concepts.							
7.	To familiarize the student with Networking concepts.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Know about Control Structures, Loop Structures and branching in programming							K3
2	Know about various searching and sorting methods.							K3
3.	Know about Functions, Recursions and Storage Classes							K3
4.	Know about Structures and Unions.							K3
5.	Know different Operating System concepts. 6.							K3
6.	Differentiate OSI Model Vs. TCP/IP suite.							K3
SYLLABUS								
UNIT-I	Review of Programming constructs Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.							
UNIT-II	Introduction to Linear Data, strings and pointers Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.							
UNIT-III	Functions, Recursions and Storage Classes Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions. Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime- memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.							

UNIT-IV	User-defined data types, Pre-processor Directives and standard storage Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming. Practice: Structure padding, user-defined data storage and retrieval programs
UNIT-V	Operating system principles and Database concepts: Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer.
Reference Books:	
1.	Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2.	The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd.
3.	Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific
4.	Object Oriented Programming in C++: N. Barkakati, PHI.
5.	Object Oriented Programming through C++ by Robat Laphore.
6.	https://www.geeksforgeeks.org/ .
7.	https://www.tutorialspoint.com/